

Natural Language Processing

Homework # 1

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21L-5692

BOS-6A

Doc ID	Text	TF	IDF	TF-IDF
1	itsy	1/4	$\log(16/4)$	0.15
	bitsy	1/4	$\log(16/4)$	0.15
	spider	1/4	$\log(16/6)$	0.10625
	water	0/4	$\log(16/2)$	0
	spout	0/4	$\log(16/4)$	0
	went	0/4	$\log(16/4)$	0
	up	0/4	$\log(16/6)$	0
4	itsy	0/5	$\log(16/4)$	0
	bitsy	0/5	$\log(16/4)$	0
	spider	1/5	$\log(16/6)$	0.085
	water	0/5	$\log(16/2)$	0
	spout	0/5	$\log(16/4)$	0
	went	0/5	$\log(16/4)$	0
	up	0/5	$\log(16/6)$	0
7	itsy	1/5	$\log(16/4)$	0.12
	bitsy	1/5	$\log(16/4)$	0.12
	Spider	1/5	$\log(16/6)$	0.085
	water	0/5	$\log(16/2)$	0
	spout	0/5	$\log(16/4)$	0
	went	0/5	$\log(16/4)$	0
	up	0/5	$\log(16/6)$	0

Cosine Similarity :-

(D1, D4)

$$\cos \theta = \frac{D1 \cdot D4}{|D1| |D4|}$$

$$D1 \cdot D4 = 0.15 \times 0 + 0.15 \times 0 + 0.10625 \times 0.085 + 0 \times 0 + 0 \times 0 + 0 \times 0 + 0 \times 0$$

$$D1 \cdot D4 = 0.00903$$

$$|D1| = \sqrt{(0.15)^2 + (0.15)^2 + (0.10625)^2 + (0)^2 + (0)^2 + (0)^2 + (0)^2}$$

$$|D1| = 0.237$$

$$|D4| = \sqrt{(0)^2 + (0)^2 + (0.085)^2 + (0)^2 + (0)^2 + (0)^2 + (0)^2}$$

$$|D4| = 0.085$$

$$\cos \theta = \frac{0.00903}{0.237 \times 0.085}$$

$$\cos \theta = 0.448$$

$$\cos \theta = 0.45$$

(D1, D7)

$$\cos \theta = \frac{D1 \cdot D7}{|D1| |D7|}$$

$$D1 \cdot D7 = 0.15 \times 0.12 + 0.15 \times 0.12 + 0.10625 \times 0.085 + 0 \times 0 + 0 \times 0 + 0 \times 0 + 0 \times 0$$

$$D1 \cdot D7 = 0.045$$

$$|D1| = \sqrt{(0.15)^2 + (0.15)^2 + (0.10625)^2 + (0)^2 + (0)^2 + (0)^2 + (0)^2}$$
$$|D1| = 0.237$$

$$|D7| = \sqrt{(0.12)^2 + (0.12)^2 + (0.085)^2 + (0)^2 + (0)^2 + (0)^2 + (0)^2}$$
$$|D7| = 0.19$$

$$\cos \theta = \frac{0.045}{0.237 \times 0.19}$$

$$\cos \theta = 0.9993$$

(D4, D7)

$$\cos \theta = \frac{D4 \cdot D7}{|D4| |D7|}$$

$$D4 \cdot D7 = 0.12 \times 0 + 0 \times 0.12 + 0.085 \times 0.085 + 0 \times 0 + 0 \times 0 + 0 \times 0 + 0 \times 0$$

$$D4 \cdot D7 = 0.007225$$

$$|D4| = \sqrt{(0)^2 + (0)^2 + (0.085)^2 + (0)^2 + (0)^2 + (0)^2 + (0)^2}$$

$$|D4| = 0.085$$

$$|D7| = \sqrt{(0.12)^2 + (0.12)^2 + (0.085)^2 + (0)^2 + (0)^2 + (0)^2 + (0)^2}$$

$$|D7| = 0.19$$

$$\cos \theta = \frac{0.007225}{0.085 \times 0.19}$$

$$\cos \theta = 0.447$$

$$\cos \theta \approx 0.45$$

The cosine similarity of (D4, D7) is maximum hence they are more similar.